Computer Programming (Monsoon '25) — Quiz 2

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Section:

Roll Number:

Do rough work on back side of pages. Assume that all the code mentioned bellow is inside the main function of a c program, with stdio.h included.

1 (2 marks)

What is the output of following program?

```c
#include <stdio.h>
void fun(int x) {
    x = 30;
}

int main() {
  int y = 20;
  fun(y);
  printf("%d", y);
  return 0;
}
```

2 (2 marks)

What is the output of following program?

```c
int main() {
    int a, *b=&a, **c=&b;
    a=4;
    **c=5;
    printf("%d", a);
}
```

3 (2 marks)

What is the output of the following program?

```c
#include <stdio.h>
#include <string.h>

int main() {
    char str[] = "abc\0def";
    printf("%s\n", str);
    printf("%d\n", strlen(str));
    printf("%d\n", sizeof(str)/sizeof(str[0]));
    return 0;
}
```

4 (3 marks)

What is the output of the following program?

```c
#include <stdio.h>
int main() {
    int arr[] = {10, 20, 30};
    int *p = arr;
    *(p + 1) = *p + 5;
    p++;
    printf("%d %d %d\n", arr[0], arr[1], *p);
    return 0;
}
```

5 **(2 marks)**

What is the output of the following program?

```c
#include <stdio.h>
int main() {
    int a = 10;
    int *p = &a;
    int **q = &p;
    int ***r = &q;
    ***r += 5;
    printf("%d %d %d\n", a, **q, ***r);
    return 0;
}
```

6 **(4 marks)**

What would be the output? Assume starting addresses, wherever required. Assume a number (assuming type conversion of %p to %d) for any address which is not specified. Clearly mention your assumption. Assume pointer takes 8 bytes.

```c
#include<stdio.h>
int main(){
    char *arr[] = {"Red","Blue","Pink",
                   "Green"};
    printf("%d\n",arr);
    printf("%d\n",arr+1);
    printf("%d\n",arr[0]);
    printf("%d\n",arr[1]);
    printf("%s\n",*arr);
    printf("%s\n",*(arr+1));
}
```

7 **(4 marks)**

What is the output?

```c
#include<stdio.h>

typedef struct Student {
  int roll_no;
  int* marks;
} Student;

int main(){
  int marks = 51;
  Student A = { 1, &marks };
  marks = 58;
  Student B = { 2, &marks };
  Student C = A;
  C.roll_no = 3;
  *(C.marks) = 60;
  printf("%d %d %d %d\n", marks, *(A.marks),
         *(B.marks), *(C.marks));
  return 0;
}
```

8 **(3 marks)**

What would be the output of the following program?

```c
#include <stdio.h>
int main() {
    int arr[] = {10, 20, 30};
    int *p = arr;
    int **q = &p;
    p++;
    printf("%d\n", **q);
    return 0;
}
```

9 **(3 marks)**

What is the output of following program?

```c
#include<stdio.h>
void f(int *p, int *q) {
  p = q;
  *p = 2;
}
int main() {
   int i = 0, j = 1;
   f(&i, &j);
   printf("%d %d", i, j);
   return 0;
}
```

10 **(3 marks)**

What is the output of following program?

```c
#include<stdio.h>
void swap (char *x, char *y) {
     char *t = x;
     x = y;
     y = t;
}

int main() {
     char *x = "Cpro";
     char *y = "Cpro25";
     char *t;
     swap(x, y);
     printf("%s, %s \n", x, y);
     t = x;
     x = y;
     y = t;
     printf("%s, %s \n", x, y);
     return 0;
}
```

11 **(3 marks)**

What would be the output of the following program?

```c
#include <stdio.h>

typedef struct {
    int id;
    int *score;
} Student;

int main() {
    int s = 90;
    Student st = {101, &s};
    Student *p = &st;
    Student **q = &p;

    (**q).score += 5;
    printf("%d \n", *((**q).score));
    return 0;
}
```

(5 marks)

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Match the following with respect to the following program segment:

int arr[3][3] = {{2,4,6}, {9,1,10}, {16, 64, 5}};

1. **arr
2. **arr <*(*arr+2)
3. *(arr[1] + 1) | arr[1][2]
4. *(arr[0]) | *(arr[2])
5. arr[0][1] ∧ arr[0][2]

i. 2

ii. 64

iii. 18

iv. 6

v. 3

vi. 0

vii. 16

viii. 1

ix. 20

x. 2

xi. 5

xii. 11

xiii. 4